

Midterm Exam Format

This is what you can expect the format of the exam to be like. This document tells you exactly the topics of each question so that you can prepare accordingly.

NOTE: This file will be further updated as the midterm is created.

1. Recursion

- Time-complexity
- Recurrence using functional notation

2. Induction Proof

- Difficulty on par with: For $n \geq 2$, show that $n^2 \geq 2n$

3. Algorithm analysis

- Function runtime analysis (tight bound using Θ)
- Understanding the average-case, worst-case and best-case for:
 - i. Discussed data structures
 - ii. Discussed algorithms
- STL sequence container operations (including amortized)
- Understanding of space complexity

4. Sorting

- Algorithms discussed in class:
 - i. Insertion Sort
 - ii. Merge Sort
 - iii. Quick Sort
 - iv. Heap Sort
 - v. Intro Sort
- Selection:
 - i. Using Heap
 - ii. Quick Select
- Implementation with different linear data structures
- Tracing passes of each algorithm (except Intro Sort)

5. BST Traversal

- Using `std::stack` and `std::heap` iteratively

6. Big-Five and Constructors

- Understanding of copy and move operations
- Knowing when each is invoked

7. AVL Trees

- No implementation details necessary
- Ability to insert and remove from an AVL Tree (and BST)
- Understanding of AVL height property and time complexity

8. Ability to use STL sequence containers to solve coding problems

- Implementation using `std::stack` and `std::queue`
- Ability to use iterators to traverse containers
- Understanding of how and why smart pointers are used

Note: You should review all quizzes, lectures, and projects to fully prepare for the exam. The topics mentioned here are not the only things you should review if you want to get a good grade. Review anything that you are not sure about from slides on the course webpage.